Data, center, and spread

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 1148 to the nearest hundred.

2. Round 11.72 to the nearest whole number.

3. Use 20 × 5 to estimate 19.8 × 5.1.

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

5. Find the range of 0, 2, 5, 10.

6. Find the MAD of 2, 5, 5, 8.

7. Data are 1, 2, 4, 4, 5, 6, 6. How many dots belong above 4 in a dot plot?

8. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

9. Order −3, 1, −6 from least to greatest.

10. Find |5| and describe it as a distance.

11. Reflect (2, -5) across the x-axis.

12. Find the distance from (-2, -3) to (-2, 2).

13. Find Q1, Q3, and IQR for 0, 2, 3, 5, 7, 9, 10, 12. Exclude the median if the count is odd.

14. Zoe averages −3, −1, 1, and 3 to find MAD. She gets 0. What should she average?

15. A $40 item is 25% off. Find the sale price.

16. Solve 3(x + 2) = 18.

17. Choose the statistical question: (A) How many pages are in one named book? (B) How many pages are in each of 13 books? Explain.

18. Make a dot plot for 6, 6, 7, 8, 8, 8, 10. State the total dots and most frequent value.

CHALLENGE / Explain your plan and check the result

C1. For 3, 5, 7, 9, 11, find the mean, each distance from the mean, and MAD. A student averages the signed changes −4, −2, 0, 2, 4 and gets 0. Explain the error and give the correct averaging calculation.

C2. A class records weekly practice hours 2, 2, 4, 6, 6, 8. If every student practices 2 extra hours next week, find the new mean, median, and range. Predict first and justify without recalculating every measure from scratch.